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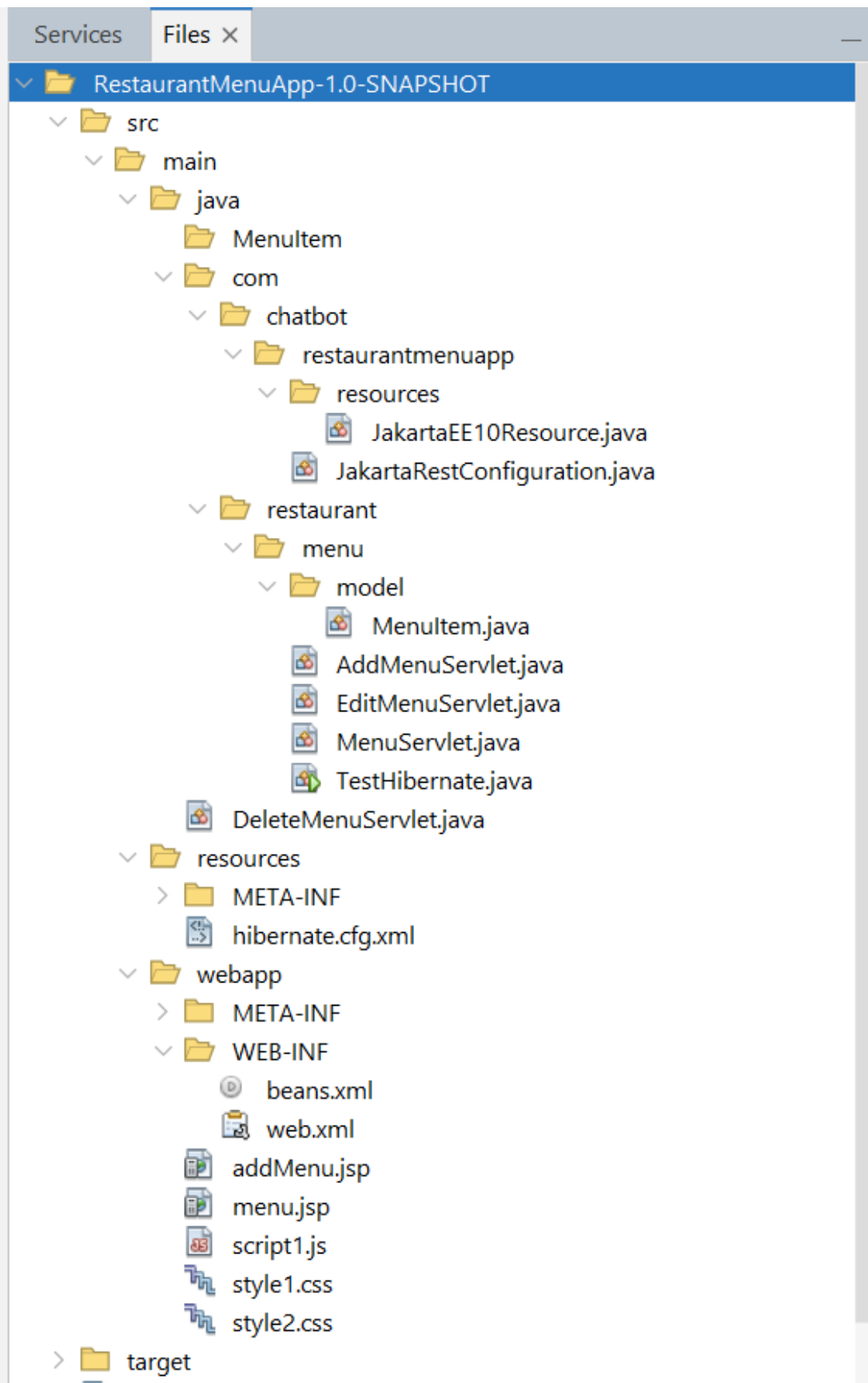
After reading documentations I have learned to use Net Beans to enhance my learning for this lab activity because in instructions its mentioned that do this using Eclipse (or any other preferred IDE) . During semester 2 I have used IntelliJ IDEA to learn advance java respectively.

Lab5 : Learning Outcomes:

Step 1: Setting Up the Maven Project

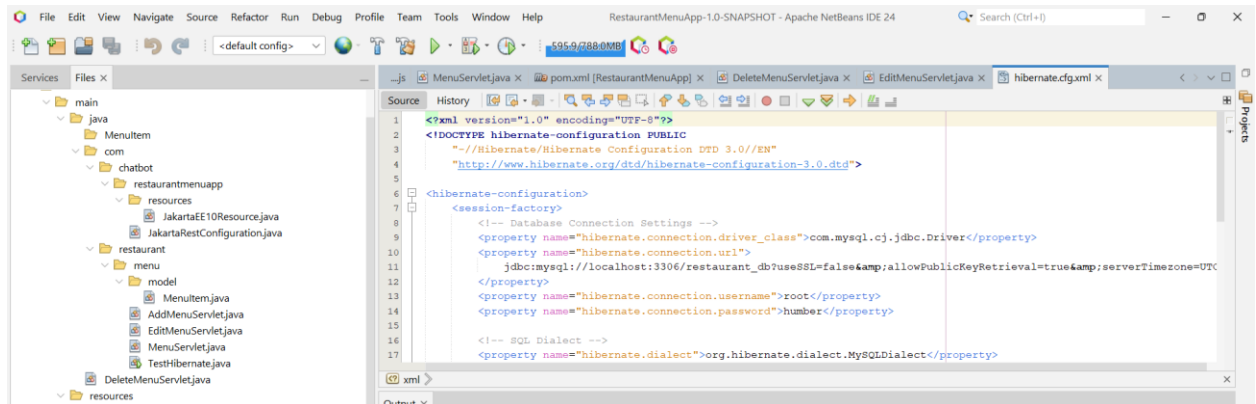
- Created a Maven project in NetBeans and configured the pom.xml file to include dependencies for Jakarta EE, Hibernate, MySQL Connector, and JSON Processing.

- This setup allowed the project to integrate Jakarta Servlets and manage database interactions through Hibernate ORM.



Step 2: Configuring Hibernate

- Added hibernate.cfg.xml in the src/main/resources directory.
- This configuration file includes database connection details (username, password, URL) and Hibernate settings for connecting to the MySQL database.



Step 3: Creating the Entity Class (MenuItem.java)

- Created the MenuItem.java class inside com.restaurant.menu.model package.
- Annotated the class with Hibernate annotations to map it to the menu_items table in the database.
- Defined fields: id, name, description, and price.

Step 4: Setting Up the Servlet to Fetch Menu Items (MenuServlet.java)

- Created MenuServlet.java in com.restaurant.menu package to handle requests for fetching menu items.
- This servlet connects to the database using Hibernate, retrieves menu items, converts them to JSON, and sends them to the front end.

Step 5: Creating the Front-End (menu.jsp)

- Created menu.jsp in the webapp folder to display the restaurant menu.
- Linked the JavaScript file (script1.js) for dynamic functionality and CSS for styling.

Step 6: Implementing JavaScript Functionality (script1.js)

Add New Menu Item

Item Name:

Oats with Fruits and nuts

Description:

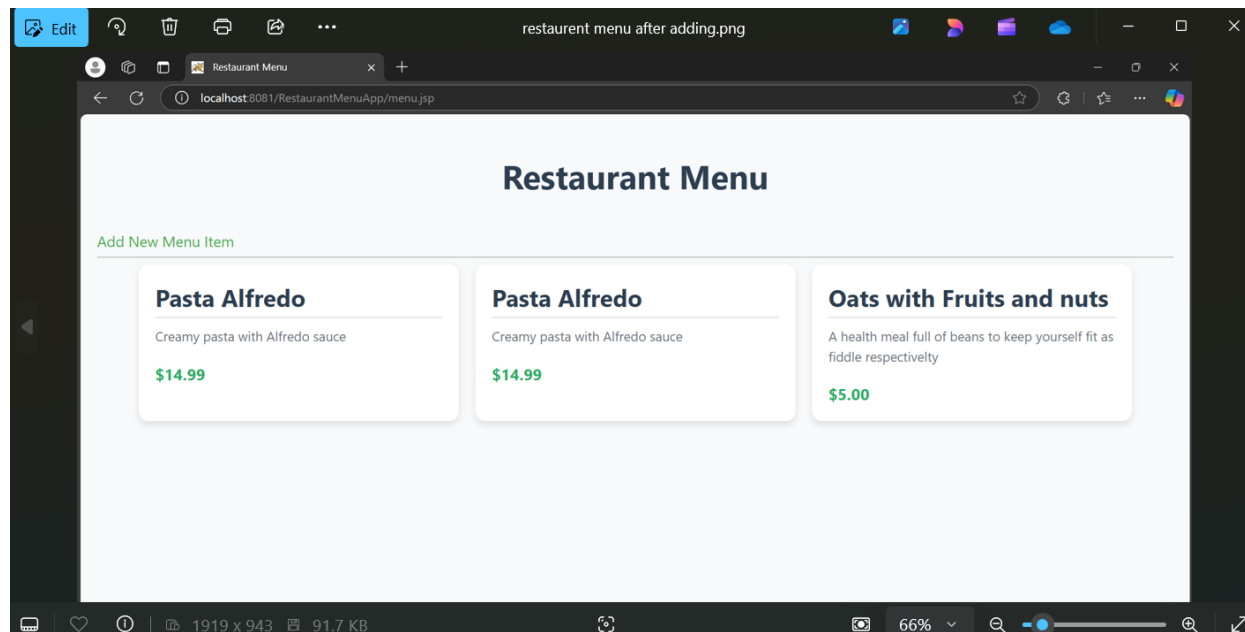
A health meal full of beans to keep yourself fit as fiddle respectvelty

Price:

5

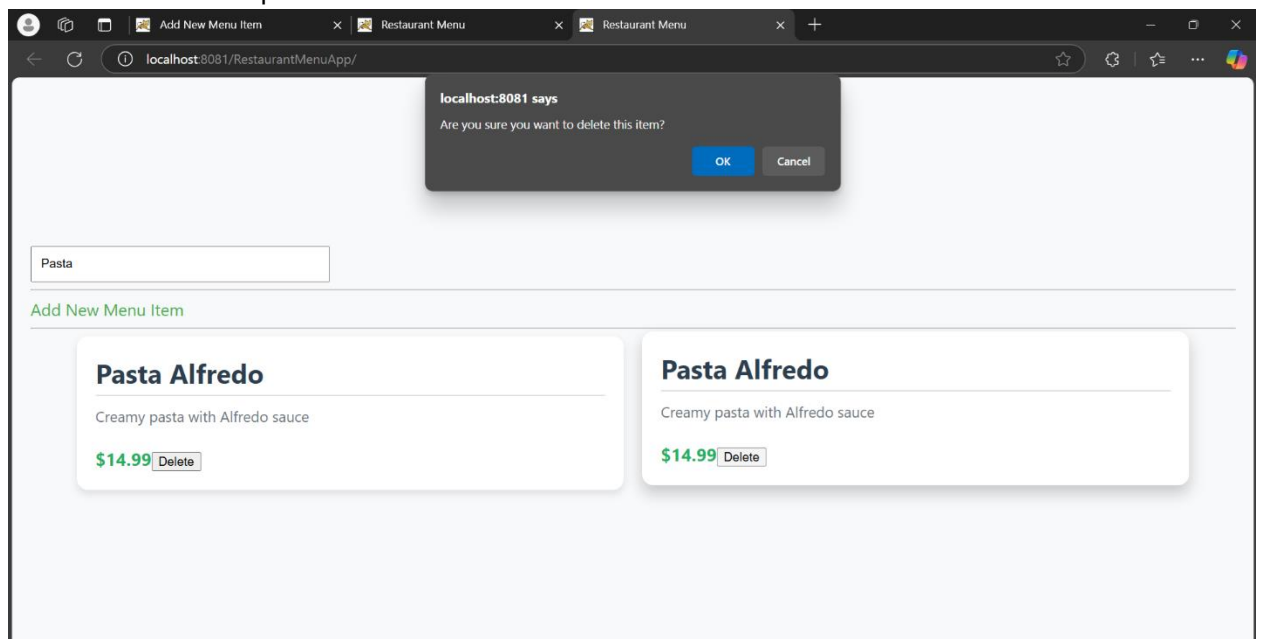
Add Item

- Added JavaScript code to:
 - Load all menu items via AJAX from MenuServlet.java.
 - Implement a search feature to filter menu items.
 - Add a "Delete" button to remove menu items via DeleteMenuServlet.java.
 - Add an "Edit" button to modify menu items via EditMenuServlet.java.



Step 7: Adding Delete Functionality (DeleteMenuServlet.java)

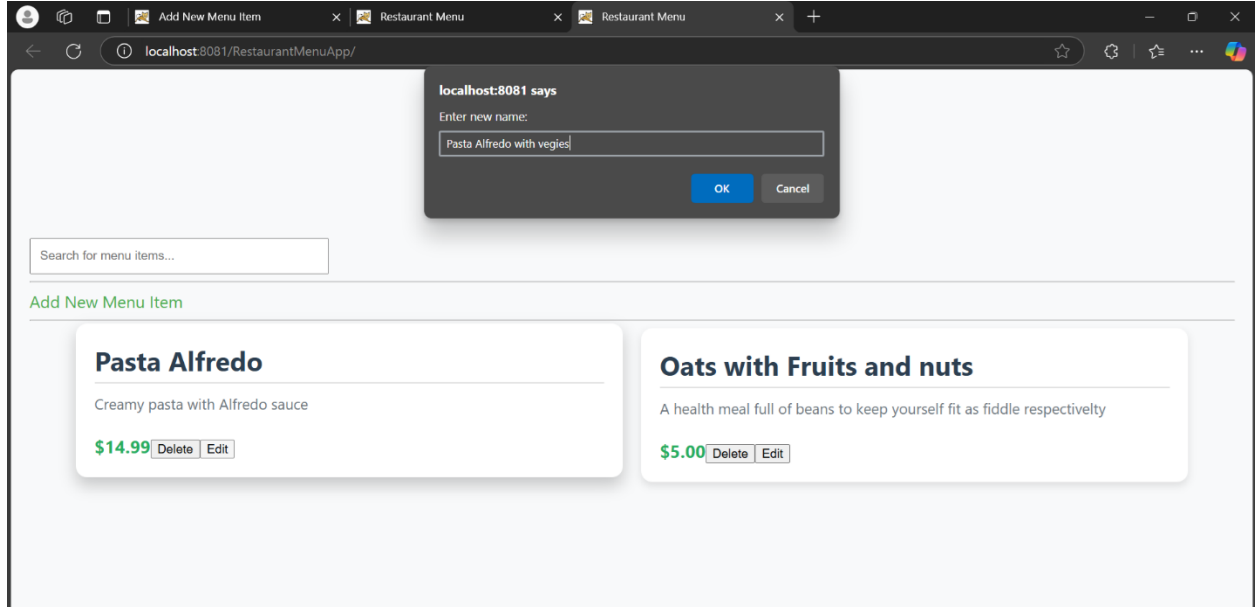
- Created DeleteMenuServlet.java to handle deletion of menu items from the database.
- The servlet receives the id of the menu item via AJAX, deletes the item from the database, and returns a success response.



Step 8: Adding Edit Functionality (EditMenuServlet.java)

- Created EditMenuServlet.java to handle updating menu items.

- The servlet receives updated name, description, and price via AJAX, updates the corresponding item in the database, and returns a success response.



Step 9: Testing and Debugging

- Cleaned and built the project to ensure all changes were compiled.
- Ran the project on Apache Tomcat and navigated to menu.jsp to verify that menu items were displayed, searchable, deletable, and editable.

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